



SANT'ANNA WORKSHOP ON CAUSAL INFERENCE

# Covariates in DiD

*OLS specs, IPW, OR, DR*

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## Recall: the $2 \times 2$ is DiD — six equivalent forms

$$1. \underbrace{\left[ \mathbb{E}[Y|D=1, t] - \mathbb{E}[Y|D=1, t-1] \right]}_{\text{first difference}} - \underbrace{\left[ \mathbb{E}[Y|D=0, t] - \mathbb{E}[Y|D=0, t-1] \right]}_{\text{first difference}}$$

$$2. \underbrace{\left[ \mathbb{E}[Y|D=1, t] - \mathbb{E}[Y|D=0, t] \right]}_{\text{group difference}} - \underbrace{\left[ \mathbb{E}[Y|D=1, t-1] - \mathbb{E}[Y|D=0, t-1] \right]}_{\text{group difference}}$$

$$3. Y = \alpha_1 + \alpha_2 \text{Post} + \alpha_3 D + \delta (\text{Post} \times D) + \varepsilon \quad \text{saturated regression}$$

$$4. Y = \alpha_1 + \tau_t + \sigma_i + \delta (\text{Post} \times D) + \varepsilon \quad \text{two-way fixed effects}$$

$$5. \Delta Y_i = \alpha_1 + \delta D_i + \varepsilon_i \quad \text{first-difference / horizontal regression}$$

$$6. \Delta \bar{Y}_G = \alpha_1 + \delta \text{Post}_G + \varepsilon_G \quad \text{group / vertical regression}$$

All six are **numerically identical**.

## The 2×2 DiD estimand

$$\underbrace{\hat{\delta}_{\text{DiD}}}_{\text{what we estimate}} = \underbrace{\text{ATT}}_{\text{what we want}} + \underbrace{\left[ \mathbb{E}[\Delta Y(0) \mid D=1] - \mathbb{E}[\Delta Y(0) \mid D=0] \right]}_{\text{PT bias}}$$

Treatment = college education. Outcome = wages.  
Parallel trends requires the PT bias term to equal zero.

## A thought experiment

| Group          | Untreated wage trend $\Delta Y(0)$ |
|----------------|------------------------------------|
| Male workers   | +10 per year                       |
| Female workers | +8 per year                        |

**Case 1:**  $D=1$  is 75% male.  $D=0$  is 75% male.

Will parallel trends hold?

## Case 1: show the work

|                                    | Calculation                      | Result |
|------------------------------------|----------------------------------|--------|
| $\mathbb{E}[\Delta Y(0) \mid D=1]$ | $0.75 \times 10 + 0.25 \times 8$ | 9.5    |
| $\mathbb{E}[\Delta Y(0) \mid D=0]$ | $0.75 \times 10 + 0.25 \times 8$ | 9.5    |

Parallel trends holds? **Yes.**

$$\hat{\delta}_{\text{DiD}} = \text{ATT} \quad \checkmark$$

## Case 2: unequal composition

|                          | % male |
|--------------------------|--------|
| $D=1$ (college educated) | 75%    |
| $D=0$ (no college)       | 25%    |

Same wage trends as before (+10 male, +8 female).

Will parallel trends hold now?

## Case 2: show the work

|                                    | Calculation                      | Result |
|------------------------------------|----------------------------------|--------|
| $\mathbb{E}[\Delta Y(0) \mid D=1]$ | $0.75 \times 10 + 0.25 \times 8$ | 9.5    |
| $\mathbb{E}[\Delta Y(0) \mid D=0]$ | $0.25 \times 10 + 0.75 \times 8$ | 8.5    |

Parallel trends holds? **No.**

$\hat{\delta}_{\text{DiD}} \neq \text{ATT}$  because PT bias =  $9.5 - 8.5 = 1.0 \neq 0$

## Why? Imbalance on a trend-causing covariate

DiD does **not** require the two groups to look the same on *levels*.

DiD **does** require the same  $Y(0)$  *trends*.

Imbalance in a covariate that *causes*  $Y(0)$  trends  
will **mechanically** break parallel trends.



## Fix: control for sex

If sex is the only covariate driving differential trends,  
then *conditioning on sex* removes the PT bias.

Several ways to do this:

**Outcome regression (OR)**   **IPW**   **Doubly robust (DR)**

## For conditioning to work: conditional PT

We need parallel trends *within* each group:

1. Male treated workers have the *same*  $Y(0)$  trend as male control workers
2. Female treated workers have the *same*  $Y(0)$  trend as female control workers

Conditional PTA:  $\mathbb{E}[\Delta Y(0) \mid D=1, X] = \mathbb{E}[\Delta Y(0) \mid D=0, X]$  for all  $X$

## The standard TWFE with additive $X$

$$Y_{it} = \alpha_i + \lambda_t + \delta (D_i \times \text{Post}_t) + \beta X_i + \varepsilon_{it}$$

Two hidden assumptions:

1. Returns to  $X$  are **constant over time**
2. Treatment effect is **the same for all  $X$**

## Worry 1: $\beta$ changes at the policy break

$$Y_{it} = \alpha_i + \lambda_t + \delta (D_i \times \text{Post}_t) + \beta X_i + \gamma (\text{Post}_t \times X_i) + \varepsilon_{it}$$

$\gamma$  absorbs any shift in returns to  $X$  between pre and post.

## Worry 2: treatment effect varies with $X$

$$Y_{it} = \alpha_i + \lambda_t + \delta_0 (D_i \times \text{Post}_t) + \tau (D_i \times \text{Post}_t \times X_i) + \beta X_i + \varepsilon_{it}$$

$$\text{ATT} = \frac{1}{n_T} \sum_{D_i=1} (\hat{\delta}_0 + \hat{\tau} X_i)$$

## Both worries: the fully saturated spec

$$Y_{it} = \alpha_i + \lambda_t + \delta_0 (D_i \times \text{Post}_t) + \boldsymbol{\tau} (D_i \times \text{Post}_t \times X_i) + \boldsymbol{\gamma} (\text{Post}_t \times X_i) + \boldsymbol{\beta} X_i + \varepsilon_{it}$$

$$\text{ATT} = \frac{1}{n_T} \sum_{D_i=1} (\hat{\delta}_0 + \hat{\boldsymbol{\tau}} X_i) \quad \cdot \quad \text{This is **Spec C** in the lab.}$$

## IPW: reweight the controls to look like the treated

$$\widehat{\text{ATT}}_{\text{IPW}} = \frac{1}{\bar{D}} \frac{1}{n} \sum_{i=1}^n \frac{D_i - \hat{p}(X_i)}{1 - \hat{p}(X_i)} \Delta Y_i$$

$$\hat{p}(X_i) = \widehat{\text{Pr}}(D_i = 1 \mid X_i) \quad (\text{propensity score}) \quad \Delta Y_i = Y_{i,\text{post}} - Y_{i,\text{pre}}$$

## IPW: five steps

1. **Estimate  $\hat{p}(X)$ .** Logit of  $D_i$  on  $X_i$ .
2. **Check overlap.** Plot  $\hat{p}$  by treatment status. Trim if needed.
3. **Construct weights.**  $w_i = (D_i - \hat{p}(X_i)) / (1 - \hat{p}(X_i))$ .
4. **Weighted average.**  $\widehat{ATT} = \sum w_i \Delta Y_i / n_T$ .
5. **Standard error.** Bootstrap or influence function.



## Overlap is non-negotiable

If  $\hat{p}(X_i) \rightarrow 1$  for a control unit, its weight  $\rightarrow \infty$ .

**No overlap = no valid counterfactual.**

This is a data problem. No estimator fixes it.

## OR: model the trend, impute the counterfactual

$$\widehat{\text{ATT}}_{\text{OR}} = \frac{1}{n_T} \sum_{D_i=1} \left[ \Delta Y_i - \hat{\mu}_0(X_i) \right]$$

$$\hat{\mu}_0(X_i) = \hat{E}[\Delta Y \mid D = 0, X] \quad \text{estimated on controls only}$$

## OR: five steps

1. **First-difference.**  $\Delta Y_i = Y_{i,\text{post}} - Y_{i,\text{pre}}$ .
2. **Fit outcome model.** Regress  $\Delta Y$  on  $X$  *using controls only*.
3. **Predict counterfactual.** Evaluate  $\hat{\mu}_0(X_i)$  for each treated unit.
4. **Mean gap.**  $\text{ATT} = \overline{\Delta Y}^T - \overline{\hat{\mu}_0(X)}^T$ .
5. **Standard error.** Bootstrap.

## DR: two chances to be right

$$\widehat{\text{ATT}}_{\text{DR}} = \frac{1}{\bar{D}} \frac{1}{n} \sum_{i=1}^n \underbrace{\frac{D_i - \hat{p}(X_i)}{1 - \hat{p}(X_i)}}_{\text{IPW correction}} \cdot \underbrace{\left[ \Delta Y_i - \hat{\mu}_0(X_i) \right]}_{\text{OR residual}}$$

DR is consistent if *either* model is right — not both

| $\hat{\mu}_0$ correct | $\hat{p}$ correct | Consistent? |
|-----------------------|-------------------|-------------|
| ✓                     | ✓                 | ✓           |
| ✓                     | ×                 | ✓           |
| ×                     | ✓                 | ✓           |
| ×                     | ×                 | <b>No</b>   |

“Doubly robust”  $\neq$  doubly unbeatable.

## Lab: ten estimators, one benchmark

Labs/Lalonde-Covariates/lalonde\_all\_specs.do

NSW treated + CPS controls · Pre = 1975, Post = 1978

RCT benchmark: **\$1,794**

Spec 0 (Naive) · Spec A (Additive  $X$ ) · Spec B (Post $\times X$ ) · Spec C (Saturated)

OR by hand · OR drdid · IPW by hand · IPW drdid · DR by hand · DR drdid

## A Monte Carlo to rank the estimators

Estimators: TWFE, OR, IPW, DR

DGPs: 4 (vary whether OR model and pscore are correct)

$n$ : 1,000    Replications: 10,000

Report: Bias, RMSE, SE, coverage, CI length

When does DR pay off? When does it break?

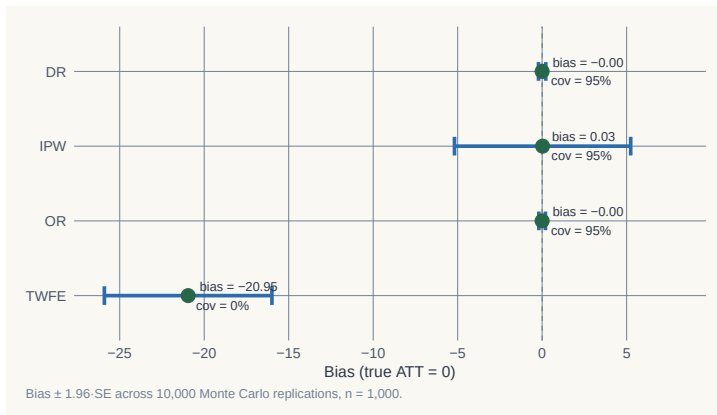
## DGP 1: both models correct

|      | <b>Bias</b> | <b>RMSE</b> | <b>SE</b> | <b>Coverage</b> | <b>CI length</b> |
|------|-------------|-------------|-----------|-----------------|------------------|
| TWFE | −20.95      | 21.12       | 2.53      | 0.000           | 9.91             |
| OR   | −0.001      | 0.10        | 0.10      | 0.950           | 0.40             |
| IPW  | +0.026      | 2.77        | 2.66      | 0.952           | 10.44            |
| DR   | −0.001      | 0.11        | 0.11      | 0.947           | 0.41             |

*TWFE: coverage zero. OR and DR: tight on truth. IPW: unbiased but wide.*



## DGP 1: the sampling distribution

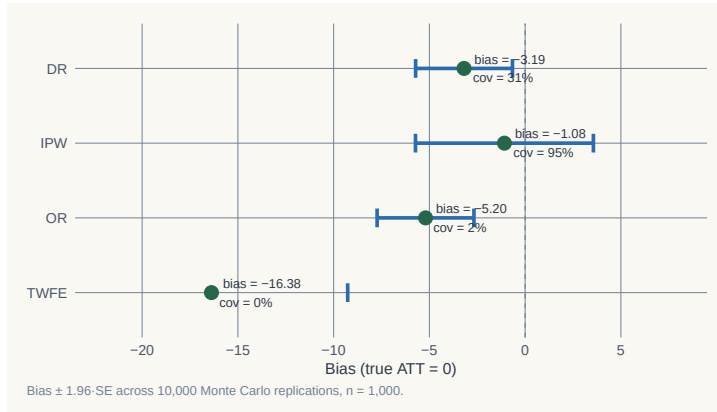


## DGP 4: both models wrong

|      | Bias   | RMSE  | SE   | Coverage | CI length |
|------|--------|-------|------|----------|-----------|
| TWFE | −16.38 | 16.54 | 3.63 | 0.000    | 14.22     |
| OR   | −5.20  | 5.36  | 1.29 | 0.015    | 5.05      |
| IPW  | −1.08  | 2.66  | 2.37 | 0.949    | 9.31      |
| DR   | −3.19  | 3.45  | 1.29 | 0.308    | 5.07      |

*DR undercovers at 0.31. IPW wins. Doubly robust  $\neq$  doubly unbeatable.*

## DGP 4: the sampling distribution



## What the Monte Carlo teaches

- **TWFE + additive  $X$** : coverage zero in every DGP. **Stop using it as a default.**
- **OR**: efficient when right. Coverage 1.5% when wrong.
- **IPW**: less efficient, but robust to outcome misspecification.
- **DR**: best when one model is right. No better than IPW when both are wrong.

## Adding $X$ doesn't make your DiD credible

Adding  $X$  shifts the credibility question—  
from **“does PT hold?”**  
to **“is my model of  $Y(0)$  or  $p(X)$  correct?”**  
You still have to answer it.

## Now let's see it with real data

LaLonde (1986) · Dehejia & Wahba (2002)

A famous dataset where we **know the true effect**.

The benchmark: **\$1,794**.

## LaLonde (1986): the RCT found $\approx$ \$800

- Job training program (National Supported Work Demonstration)
- Randomized — treated and control groups assigned by lottery
- RCT estimate: earnings effect  $\approx$  **\$800**

A clean causal estimate. The benchmark exists.

## LaLonde: dropped the RCT controls, used CPS instead

- Replaced the randomized controls with observational data (CPS, PSID)
- Re-ran every evaluator method from the literature
- Results ranged from **−\$16,000 to +\$3,000**

The true effect was \$800. Not one method recovered it.

*“The available evidence calls into serious question the ability of nonexperimental evaluators. . . ”*

Empirical labor economics faces an evaluation problem.



## Dehejia & Wahba (2002): propensity scores get close

- Need **two pre-treatment years** for all datasets — uses a shorter sample
- RCT benchmark on this shorter sample: **\$1,794**
- Applied propensity score matching on the non-experimental sample
- Result: estimates close to \$1,794

Selective use of the data + right method = nearly recovered the truth.

## Our exercise: DiD with and without covariates

NSW treated + CPS controls · Pre = 1975 · Post = 1978

We know the effect: **\$1,794**

Let's see which specs recover it — and which don't.

## Naive TWFE: \$3,621 — almost double the truth

```
reg re i.post##i.ever_treated, robust
```

| Estimator  | ATT            | vs benchmark    |
|------------|----------------|-----------------|
| Naive TWFE | <b>\$3,621</b> | <b>+\$1,827</b> |

No covariates — the CPS controls trend very differently from NSW treated.

## Additive $X$ : still \$3,621

```
reg re i.post##i.ever_treated $X, robust
```

| Estimator    | ATT            | vs benchmark    |
|--------------|----------------|-----------------|
| Additive $X$ | <b>\$3,621</b> | <b>+\$1,827</b> |

Adding  $X$  as time-invariant regressors does nothing here. The unit FE absorbs the level differences. The trend bias remains.

## Post $\times$ X: \$1,711 — now in the right range

```
reg re i.post##i.ever_treated ///
    i.post##c.age i.post##c.agesq i.post##c.agecube ///
    i.post##c.educ i.post##c.educsq ///
    i.post##i.marr i.post##i.nodegree ///
    i.post##i.black i.post##i.hisp ///
    i.post##c.re74 i.post##i.u74, robust
```

| Estimator       | ATT            | vs benchmark |
|-----------------|----------------|--------------|
| Post $\times$ X | <b>\$1,711</b> | −\$83        |

## Saturated TWFE: \$1,770

```
* First-difference, then regress dy on X + D x X interactions
reg dy $X ///
    i.ever_treated#c.age i.ever_treated#c.agesq ///
    i.ever_treated#c.educ i.ever_treated#c.re74 ///
    i.ever_treated, robust
* Predict counterfactuals; ATT = mean(tau_sat) for treated
```

| Estimator      | ATT            | vs benchmark |
|----------------|----------------|--------------|
| Saturated TWFE | <b>\$1,770</b> | −\$24        |

## OR (HIT 1997): same answer as saturated TWFE

```
* Fit trend model on controls only
reg dy $X if ever_treated == 0, robust
predict dy_hat
* ATT = mean gap for treated
gen delta_hat = dy - dy_hat if ever_treated == 1
```

| Estimator              | ATT            | vs benchmark |
|------------------------|----------------|--------------|
| OR by hand / drdid reg | <b>\$1,770</b> | −\$24        |

Saturated TWFE and OR are algebraically equivalent in the 2-period case.

## IPW (Abadie 2005): \$1,861

```
drdid re $X, time(year) ivar(id) tr(ever_treated) ipw
```

| Estimator | ATT            | vs benchmark |
|-----------|----------------|--------------|
| IPW       | <b>\$1,861</b> | +\$67        |

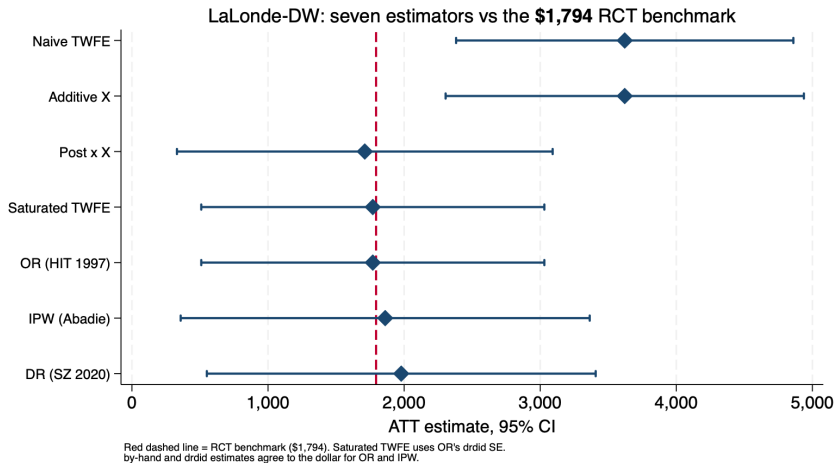


## DR (Sant'Anna–Zhao 2020): \$1,979

```
drdid re $X, time(year) ivar(id) tr(ever_treated) dripw
```

| Estimator | ATT            | vs benchmark |
|-----------|----------------|--------------|
| DR        | <b>\$1,979</b> | +\$185       |

## All seven estimators vs the \$1,794 benchmark



## Covariates are not robustness checks — they are design

- People say: *“if adding covariates changes results, be suspicious.”*
- That gets it backwards.
- We **know** the effect is \$1,794.
- Without covariates, or with additive covariates, we get \$3,621 — *wrong*.
- The correct specs recover \$1,770–\$1,979 — *right*.

Covariates fix broken parallel trends. Changing your result is not a bug. It is the design working.

The question is always  
**“did I get the model right?”**

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